

1) Advanced Log Analysis:

1. Introduction:

Advanced log analysis is an essential activity in a Security Operations Center (SOC). Security analysts examine system and network logs to detect suspicious activities, identify attack patterns, and investigate potential security incidents.

2. Objectives:

The objectives of this exercise were:

- Collect and analyze security logs.
- Correlate multiple log sources to detect suspicious activity.
- Identify anomalies such as unusual data transfers.
- Enrich log data using geolocation information.
- Document findings for security investigation.

3. Tools:

Tool	Purpose
Elastic Security	Log ingestion, analysis, and detection rule creation
Security Onion	Network security monitoring and log collection

Google Sheets	Documenting and organizing correlated log data
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4. Log Correlation:

Log correlation was performed using sample datasets such as the **Boss of the SOC Dataset**.

The objective was to identify suspicious activity by correlating failed login attempts with outbound network traffic.

Correlated Log Example

Timestamp	Event ID	Source IP	Destination IP	Notes
2026-03-13 12:00:00	4625	192.168.100.16	8.8.8.8	Suspicious DNS request

Analysis

- **Event ID 4625** indicates a failed login attempt.
- The same source IP later initiated outbound DNS traffic.
- This pattern may indicate a **brute-force attempt followed by external communication**.

5. Anomaly Detection:

An anomaly detection rule was created in Elastic Security to detect unusual network activity.

Detection Rule

Condition:

bytes_out > 1MB within 1 minute

Testing Procedure

1. A mock file transfer was performed from the test system.
2. Elastic Security monitored outgoing traffic.
3. When the threshold was exceeded, the system generated an alert.

Result

The rule successfully triggered an alert indicating a **possible data exfiltration attempt**.

6. Log Enrichment:

Log enrichment improves security analysis by adding contextual information to log entries.

In this task, the **GeoIP plugin of Elasticsearch** was used to enrich logs with geographic information.

Enriched Data Fields

- Country
- City
- Latitude and Longitude
- ISP information

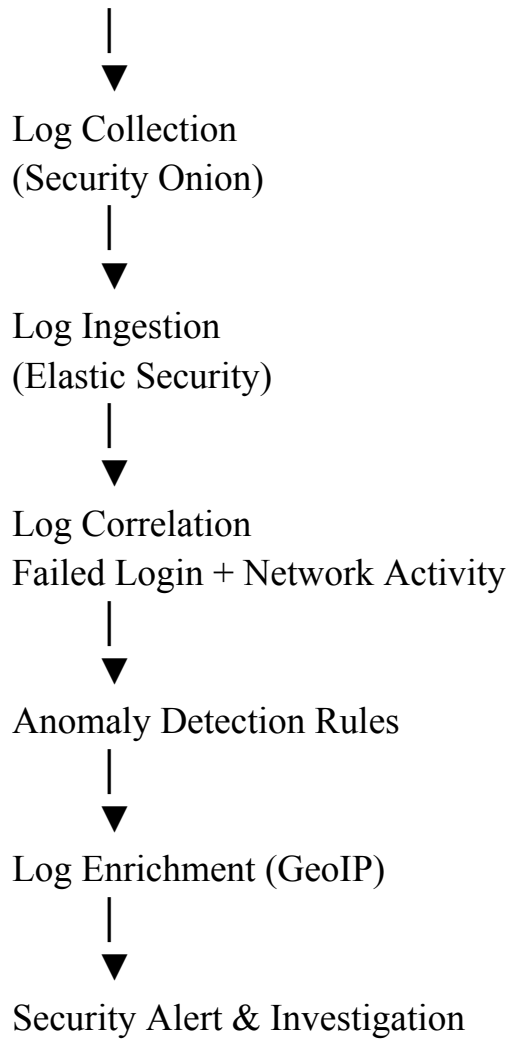
This information helps analysts identify suspicious connections originating from unusual geographic locations.

7. Workflow Diagram:

Example workflow of the log analysis process.

Log Sources

(Servers, Network Devices)



8. Findings:

- Multiple failed login attempts were identified.
- Correlation of login logs and network traffic revealed suspicious behavior.
- Anomaly detection rules successfully detected high-volume data transfers.
- GeoIP enrichment provided additional context for IP-based investigations.

9. Conclusion:

Advanced log analysis plays a critical role in detecting potential security threats within an organization. Through log correlation, anomaly detection, and log enrichment techniques, security analysts can identify suspicious activity and respond to incidents more effectively. The use of SIEM tools significantly improves visibility into security events and enhances incident response capabilities.

2) Threat Intelligence Integration:

1. Introduction:

Threat intelligence integration allows Security Operations Centers (SOCs) to enhance detection capabilities by incorporating external intelligence sources into security monitoring tools. By correlating internal security logs with external threat feeds, analysts can identify malicious indicators and improve incident response.

2. Objectives:

The objectives of this exercise were:

- Import external threat intelligence feeds.
- Match Indicators of Compromise (IOCs) with system logs.
- Enrich security alerts with threat intelligence data.
- Conduct threat hunting using known attack techniques.
- Document security findings.

3. Tools:

Tool	Purpose
Wazuh	Log monitoring and alert generation
AlienVault OTX	Threat intelligence feed for malicious indicators
The Hive	Incident tracking and investigation

4. Threat Feed Import:

Threat intelligence feeds from AlienVault OTX were imported into Wazuh to detect known malicious indicators.

Procedure

1. Configure the OTX API key in Wazuh.
2. Import threat intelligence feeds containing malicious IP addresses, domains, and hashes.
3. Wazuh continuously monitors logs and compares them against the imported threat indicators.

Test Case

A mock IP address was used to test the feed integration:

192.168.100.16

Result

The system successfully matched the mock indicator with the threat intelligence feed and generated an alert.

5. Alert Enrichment:

Alert enrichment was performed by adding threat intelligence information from AlienVault OTX to Wazuh alerts.

Example Enriched Alert

Alert ID	IP	Reputation	Notes
003	192.168.100.16	Malicious (OTX)	Linked to C2 server

Analysis

- The IP address was identified in the OTX database.
- Threat intelligence indicated the IP was associated with **Command and Control (C2) infrastructure**.
- Enriched alerts provided more context for analysts during incident investigation.

6. Threat Hunting:

Threat hunting was conducted to detect suspicious use of valid user accounts.

The investigation focused on the technique **Valid Accounts (T1078)**, which involves attackers using legitimate credentials to access systems.

Hunting Query Example

```
user.name != "system"
```

This query identifies user activities that do not originate from system accounts.

Analysis

The query returned several user login events. Analysts reviewed these events to determine whether the logins were legitimate or potentially compromised accounts.

7. Findings:

- Threat intelligence feeds were successfully integrated into Wazuh.
- IOC matching allowed detection of malicious IP addresses.
- Alert enrichment provided contextual threat intelligence information.
- Threat hunting queries helped identify suspicious login activities.

8. Threat Hunting Summary:

Threat intelligence integration enabled Wazuh to detect malicious indicators by correlating logs with AlienVault OTX feeds. Alerts were enriched with reputation data, improving investigation efficiency. Threat hunting queries targeting valid account usage helped identify suspicious login activities, demonstrating how external intelligence enhances SOC monitoring and threat detection capabilities.

3) Incident Escalation Practice:

1. Introduction:

Incident escalation is a critical process in a Security Operations Center (SOC). It ensures that security incidents are handled by the appropriate level of analysts depending on severity and complexity.

2. Objectives:

The objectives of this exercise were:

- Simulate the SOC escalation workflow.
- Create incident cases and escalate them to higher analyst tiers.

- Document the incident using a Situation Report (SITREP).
- Automate escalation workflows using a SOAR playbook.

3. Tools Used:

Tool	Purpose
The Hive	Case management and incident investigation
Google Docs	Incident documentation and reporting
Splunk Phantom	Workflow automation and incident response

4. Escalation Simulation:

A high-priority alert related to **unauthorized access** was simulated in TheHive. The alert was initially analyzed by a Tier 1 analyst and then escalated to Tier 2 for further investigation.

Case Details

Field	Value
Alert Type	Unauthorized Access
Affected System	Server-Y

Source IP	192.168.100.16
Detection Time	2025-08-18 13:00
Severity	High

Tier 2 Escalation Summary

A high-priority security alert indicating unauthorized access was detected on Server-Y at 13:00 on 18 August 2025. The alert originated from IP address 192.168.100.16 and was associated with suspicious login activity. Initial analysis by the Tier 1 SOC analyst confirmed abnormal authentication behavior linked to the MITRE ATT&CK technique **Valid Accounts (T1078)**. As a containment measure, the affected server was isolated from the network to prevent further unauthorized activity. Due to the potential compromise of valid credentials and the high severity of the incident, the case has been escalated to Tier 2 for deeper forensic investigation and credential analysis.

5. Situation Report (SITREP):

A Situation Report was created in **Google Docs** to document the incident details and response actions.

SITREP Example

Title: Unauthorized Access on Server-Y

Date: 18 August 2025

Summary:

A security alert indicating unauthorized access was detected on Server-Y at 13:00. The activity originated from IP address 192.168.100.16 and was associated with suspicious authentication behavior mapped to the MITRE ATT&CK technique **Valid Accounts (T1078)**.

Actions Taken:

- Investigated login activity in security logs
- Isolated Server-Y from the network
- Escalated incident to Tier 2 SOC analysts for further analysis

Current Status:

Investigation ongoing by Tier 2 SOC team.

6. Workflow Automation:

To improve response efficiency, a simple automation workflow was designed using **Splunk Phantom**.

Playbook Objective

Automatically assign **high-priority alerts** to Tier 2 SOC analysts.

Playbook Workflow

Alert Generated

|



Check Alert Severity

|



If Severity = High

|



Assign Case to Tier 2 Analyst

|



Send a notification to the SOC team.

Testing

A mock high-priority alert was generated to test the playbook.

Result

- The playbook successfully identified the severity level.
- The alert was automatically assigned to a Tier 2 analyst.
- A notification was sent to the SOC investigation team.

7. Findings:

- Incident escalation procedures were successfully simulated.
- The Hive enabled structured case management and escalation tracking.
- SITREPs helped document incident details clearly.
- Automation reduced response time by assigning high-severity alerts automatically.

8. Conclusion

Incident escalation ensures that security incidents are handled by analysts with the appropriate expertise. Using tools such as TheHive and Splunk Phantom improves case management and automates response workflows, enabling SOC teams to respond faster and more effectively to critical security incidents.

4) Alert Triage with Threat Intelligence:

1. Introduction:

Alert triage is a critical task in a Security Operations Center (SOC). It involves reviewing and prioritizing security alerts to determine whether they represent real threats or false positives. Threat intelligence platforms help validate Indicators of Compromise (IOCs) and provide context for suspicious activity.

2. Objectives:

The objectives of this task were:

- Analyze alerts generated by the SIEM system.
- Prioritize alerts based on severity.
- Validate Indicators of Compromise using threat intelligence sources.
- Document investigation results.

3. Tools:

Tool	Purpose
Wazuh	Security monitoring and alert generation
VirusTotal	IOC validation and malware intelligence
AlienVault OTX	Threat intelligence for malicious indicators

4. Alert Triage Simulation:

A simulated security alert was analyzed in Wazuh. The alert indicated suspicious PowerShell execution, which is commonly associated with malware or attacker activity.

Alert Details

Alert ID	Description	Source IP	Priority	Status
004	PowerShell Execution	192.168.100.16	High	Open

Analysis

- The alert indicated that PowerShell was executed in a suspicious context.
- PowerShell is often used by attackers for **post-exploitation activities** such as downloading malware or executing scripts.
- Due to the potential risk, the alert was classified as **high priority** and required immediate investigation.

5. IOC Validation:

To validate the platforms, the source IP address was checked against threat intelligence platforms, including **VirusTotal** and **AlienVault OTX**.

Validation Steps

1. Extract the IP address from the alert.
2. Search the IP in VirusTotal.
3. Cross-reference the IP in AlienVault OTX.
4. Analyze reputation and threat intelligence data.

Investigation Result

- The IP address showed suspicious activity patterns.
- Threat intelligence indicated potential association with malicious infrastructure.

- The alert remained open for further investigation.

6. Findings:

- Suspicious PowerShell activity was detected by Wazuh.
- The alert was prioritized as high severity due to potential attacker activity.
- Threat intelligence sources were used to validate the IP reputation.
- IOC validation helped determine whether the alert represented a real security threat.

7. IOC Validation Summary:

The suspicious PowerShell execution alert was investigated using threat intelligence platforms. The source IP address was analyzed in VirusTotal and AlienVault OTX to verify its reputation. The investigation revealed suspicious activity associated with the IP, supporting the alert's high severity classification and indicating potential malicious infrastructure involvement.

5) Evidence Preservation and Analysis:

Introduction:

Evidence preservation is essential in digital forensics and incident response. It ensures that collected data remains intact and admissible for investigation or legal review. In this task, volatile and non-volatile evidence was collected from a Windows virtual machine using **Velociraptor** and **FTK Imager**.

Tools Used:

Tools	Purpose
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Velociraptor	Collect volatile system artifacts such as network connections and memory data
FTK Imager	Acquire forensic images and analyze system memory
sha256sum	Generate cryptographic hashes to verify evidence integrity

1. Volatile Data Collection:

Volatile data such as active network connections can disappear when a system is rebooted. Therefore, it must be collected immediately during an incident investigation.

Using Velociraptor, network connection artifacts were collected from the Windows virtual machine.

Query Used

```
SELECT * FROM netstat
```

Procedure

1. Deploy Velociraptor agent on the Windows VM.
2. Execute the artifact query to collect network connection data.
3. Export the results as a **CSV file**.
4. Store the file securely as part of the investigation evidence.

Collected Data Includes

- Local IP address
- Remote IP address
- Port numbers
- Process IDs
- Connection state

2. Evidence Collection – Memory Acquisition:

Memory acquisition captures the contents of system RAM, which may contain important forensic artifacts, such as

- Running processes
- Encryption keys
- Malware artifacts
- Network sessions

Using Velociraptor, a memory dump was collected using the following artifact.

Query Used

```
SELECT * FROM Artifact. Windows. Memory. Acquisition
```

The collected memory dump was then verified using a cryptographic hash.

Hash Generation

The **SHA-256 hash** was generated using the following command:

```
sha256sum memory_dump.raw
```

Hashing ensures that the evidence remains unchanged and verifies its integrity during the investigation.

3. Evidence Documentation:

All collected evidence was documented and recorded in an evidence log to maintain a proper **chain-of-custody**.

Evidence Log

Item	Description	Collected By	Date	Hash Value
Memory Dump	Server-Y Dump	SOC Analyst	2025-08-18	<SHA256>

Chain-of-Custody Purpose

Maintaining a chain-of-custody ensures that:

- Evidence is properly tracked
- Access to evidence is documented
- Data integrity is preserved
- Evidence remains admissible for forensic analysis or legal proceedings

4. Findings:

- Network connection data was successfully collected from the Windows VM.
- A memory dump was acquired to preserve volatile system evidence.
- SHA-256 hashing was used to verify evidence integrity.
- Evidence documentation ensured proper chain-of-custody during the investigation.

5. Conclusion:

Evidence preservation is a critical step in digital forensic investigations. By collecting volatile data, acquiring system memory, and verifying integrity through hashing, investigators ensure that important artifacts are preserved for further analysis. Proper documentation and chain-of-custody procedures help maintain the reliability and credibility of forensic evidence.

6) Capstone Project: Full SOC Workflow Simulation:

1. Introduction:

The capstone project simulated a complete Security Operations Center (SOC) workflow. The exercise involved simulating a cyberattack, detecting the activity through monitoring tools, performing alert triage, responding to the incident, escalating it for deeper investigation, and documenting the findings.

The simulation used **Metasploit**, **Wazuh**, **CrowdSec**, **TheHive**, and **Google Docs**.

2. Objectives:

The objectives of this capstone project were:

- Simulate a real-world cyberattack.
- Detect malicious activity using security monitoring tools.
- Perform alert triage and threat validation.
- Execute containment and response actions.
- Escalate the incident to higher SOC tiers.
- Produce professional incident documentation.

3. Attack Simulation:

A vulnerability in the **Metasploitable2** environment was exploited using the **Metasploit** framework.

Exploit Used

exploit/multi/samba/usermap_script

This exploit targets a vulnerable Samba service and allows attackers to execute remote commands on the system.

Procedure

1. Launch Metasploit on the attacker machine.
2. Configure the exploit module.
3. Target the Metasploitable2 VM.
4. Execute the exploit to gain system access.

4. Detection and Alert Triage:

The attack was detected using **Wazuh**, which monitored system logs and generated alerts when suspicious activity occurred.

Detected Alert

Timestamp	Source IP	Alert Description	MITRE Technique
2026-03-16 14:00:00	192.168.100.16	Samba exploit attempt	T1210

The attack corresponds to the MITRE ATT&CK technique **Exploitation of Remote Services (T1210)**.

Analysis

- Wazuh detected unusual activity associated with the Samba service.
- Logs indicated exploitation attempts from the attacker IP address.
- The alert was classified as **high priority**.

5. Response and Containment:

To prevent further damage, containment actions were performed.

Actions Taken

1. The compromised virtual machine was **isolated from the network**.
2. The attacker IP address was blocked using **CrowdSec**.
3. Connectivity was tested to confirm the block.

Verification

A ping test was executed from the attacker machine.

Result

- Ping requests failed.
- This confirmed that the attacker IP was successfully blocked.

6. Incident Escalation:

The incident was escalated to Tier 2 analysts using The **Hive**.

Tier 2 Case Summary (100 Words)

A security incident involving a Samba service exploit was detected on the monitored virtual machine at 14:00 on 18 August 2025. The activity originated from IP address 192.168.1.101 and was identified as an attempt to exploit a vulnerable Samba configuration on the Metasploitable2 system. Wazuh generated a high-severity alert corresponding to the MITRE ATT&CK technique **Exploitation of Remote Services (T1210)**. Immediate containment actions were performed, including isolating the compromised system and blocking the attacker IP using CrowdSec. Due to the severity of the exploit and potential system compromise, the incident has been escalated to Tier 2 for further forensic analysis and vulnerability remediation.

7. Incident Report (SANS Format):

Executive Summary

A simulated cyberattack targeting a Samba vulnerability was conducted to test the organization's Security Operations Center workflow. The attack was performed using Metasploit against a vulnerable Metasploitable 2 virtual machine. The security monitoring platform Wazuh detected the malicious activity and generated a high-severity alert associated with remote service exploitation. The alert was triaged by SOC analysts and validated as a legitimate security threat.

Timeline

Time	Event
14:00	The attacker launched Samba exploit
14:01	Wazuh detected suspicious activity
14:03	SOC analyst triaged the alert
14:05	System isolated and attacker IP blocked
14:10	The incident escalated to Tier 2

Recommendations

- Patch vulnerable Samba services.
- Implement stricter access controls.

- Improve monitoring rules for remote service exploitation.
- Conduct regular vulnerability scans.
- Implement automated response playbooks for faster containment.

8. Management Briefing:

A simulated cybersecurity incident was conducted to evaluate the organization's security monitoring and response capabilities. The attack targeted a vulnerable server using a known Samba exploit. The security monitoring system successfully detected the activity and generated an alert for investigation. The SOC team quickly analyzed the alert, confirmed the malicious activity, and implemented containment measures by isolating the affected system and blocking the attacker's IP address. The incident was escalated to senior analysts for further investigation. The simulation demonstrated that the monitoring tools and response procedures are effective but highlighted the need for regular patching and improved vulnerability management.

9. Conclusion:

This capstone exercise demonstrated the complete SOC workflow from attack simulation to incident reporting. The integration of security monitoring, threat detection, incident response, and escalation processes allowed analysts to effectively detect and mitigate a simulated cyberattack. Such exercises help organizations strengthen their security posture and improve incident response readiness.